Wenkai Zhu

3D Vision | Reliable Multimodal Perception | Autonomous Systems

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RESEARCH PROFILE

Master's candidate at Southeast University researching reliability-aware multimodal 3D perception and sensor fusion under imperfect observations. My work spans LiDAR-camera fusion, stereo vision and 3D perception. Through research and engineering projects, I have also worked with real vehicle platforms, sensor data collection and deployment, and I am interested in building autonomous perception systems that remain reliable in practice.

EDUCATION

Southeast University Sep 2024 - Jun 2027 (expected)

School of Mechanical Engineering | Department of Vehicle Engineering

Professional master's candidate; Major: Machinery

Advisors: Prof. Guodong Yin and Assoc. Prof. Keke Geng

China University of Mining and Technology Sep 2020 - Jun 2024

School of Mechatronic Engineering | Bachelor's degree in Mechanical Engineering

RESEARCH OUTPUTS

- **Wenkai Zhu**, Keke Geng, Zhichao Liu, Hansen Wang. HC-MVMM: Occlusion-Aware Hierarchical Confidence Modeling for LiDAR-Camera 3D Object Detection. **BMVC 2026 (accepted).**
  Front-surface colouring and point-voxel-BEV confidence propagation mitigate occlusion-induced colour contamination.
- Wenkai Zhu, et al. RCHead: Reliability-Calibrated Ranking and Refinement for Center-Based 3D Object Detection. Manuscript in preparation.
  Jointly models IoU quality and structured reliability for score calibration and residual box refinement.
- Keke Geng, Wenkai Zhu, Guodong Yin. Method and System for LiDAR-Camera 3D Object Detection Guided by Occlusion Awareness and Hierarchical Confidence.
  Chinese invention patent application CN122618234A, published; second inventor.

RESEARCH & ENGINEERING EXPERIENCE

Roadside Multimodal Perception | Core Developer Jan 2024 - Dec 2026 (project term)

Yangtze River Delta Science and Technology Innovation Joint Project

- Led LiDAR-camera 3D detection research and deployment, producing HC-MVMM and the RCHead manuscript.
- Integrated radar for target association, state estimation and tracking; built a UDP vehicle-infrastructure prototype.

Stereo Road Preview System | Core Developer Jul 2025 - Present

Xiamen King Long Motor Group Collaboration

- Designed synchronized ground-truth collection using dual industrial cameras and two LiDAR sensors; performed point-cloud registration and temporal accumulation to build a multi-scene road-surface perception dataset.
- Developed HITNet-based stereo depth, pseudo-point-cloud reconstruction and road-elevation estimation; added depth-gated 3D tracking and ROS/RViz visualization, with real-time Jetson Orin deployment (~15 FPS).

FMCW LiDAR-Based AEB Perception | Core Algorithm Developer Feb 2025 - Jun 2025

Skywell New Energy Vehicles Group Collaboration

- Developed velocity-aware point-cloud preprocessing, clustering, association and motion prediction; optimised grid-based clustering and velocity-preserving downsampling to achieve ~50 ms end-to-end latency.

Human-AI Authority Allocation | Algorithm & Software Developer Oct 2024 - Nov 2025

China Aeronautical Radio Electronics Research Institute Collaboration

- Implemented the C++ neural-network model, user interface and UDP communications; integrated the system with a cockpit task-network simulator.

Multi-Platform Remote Sensing | Proposal Contributor Feb 2025 - Apr 2025

- Contributed to proposal design for UAV-satellite-cloud collaborative sensing, covering multi-source detection, robust feature extraction, temporal change detection and lightweight models.

SKILLS & HONOURS

Methods: multisensor fusion, reliability modelling, stereo depth, point-cloud processing, multi-object tracking.
Tools: Python, C++, PyTorch, OpenCV, PCL, Linux/CUDA, ROS/ROS2, RViz, Git; Jetson deployment.
Honours: Southeast University Second-Class Graduate Scholarship (2025); National Third Prize, 19th National Student Intelligent Car Competition, Outdoor 5G Remote-Driving Autonomous Vehicle (2024).